Dairu Liu

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EDUCATION

Nankai University Tianjin, China
B.Eng. in Software Engineering; GPA: 3.75; Top 15% *Sep. 2023 – Jun. 2027 (expected)*

The Affiliated High School of Peking University Beijing, China

RESEARCH INTERESTS

My primary research goal is to leverage large-scale data to build **generalizable** embodied agents capable of performing diverse real world tasks. I am particularly interested in **Robot Learning** (scaling up robot data, efficient robot learning algorithms, humanoid whole-body control, loco-manipulation) and **Multimodal Reasoning** (agentic perception, cross-modal understanding, long-horizon decision making).

PUBLICATIONS AND PREPRINTS

[1] **HumanTracker: Towards Comprehensive and Human-Aligned Motion Tracking Benchmark**
Dairu Liu*, Zekun Qi*, Jiayu Zeng*, Yu Guan, Chenghuai Lin, Xuchuan Chen, Xinqiang Yu, Wenyao Zhang, He Wang†, Li Yi†
ECCV 2026 (Score: 4.33/6.0; Top 2% Submission)

- Built a standardized humanoid motion tracking benchmark with **150 hours** of newly captured optical mocap from **24 professional performers**, comprising **24K clips** organized into **4 motion families** for fine-grained diagnosis.
- Proposed **HumanScore**, a preference-aligned reward model trained on **3K pairwise human comparisons**; achieves **0.8834** alignment rate with human judgments, outperforming MPJPE (**0.8164**), and validated on **Unitree G1** real-robot deployment.

[2] **Thinking with Visual Abstract: Enhancing Multimodal Reasoning via Visual Abstraction** **11 citations**
Dairu Liu*, Ziyue Wang*, Minyuan Ruan*, Fuwen Luo, Chi Chen, Peng Li†, Yang Liu†
COLM 2026 Under Review
NeurIPS 2025 Workshop [Paper] [Project]

- Proposed **Visual Abstract Thinking** (**VAT**), a reasoning paradigm that replaces verbose verbal CoT with visual abstractions, focusing models on semantic, structural, and geometric cues.
- VAT achieves +**2.21%** average gain over the GPT-5 baseline, exceeding CoT (+**0.96%**); uses only **0.68**× CoT runtime and **0.1**× the tokens of Visual Sketchpad on GPT-4o. RL-trained VAT further gains +**4.52%** on Qwen-2.5-VL-3B.

[3] **LIMMT: Less is More for Motion Tracking**
Yu Guan*, Zekun Qi*, Chenghuai Lin, ***Dairu Liu***, Xuchuan Chen, Wenyao Zhang, Jilong Wang, Xinqiang Yu, He Wang†, Li Yi†
ICML 2026

- First data centric study for humanoid motion tracking: **General Quality Selection** (**GQS**) scores motion along physics feasibility, diversity (HME embeddings), and complexity, then selects compact subsets via complexity-weighted farthest-point sampling.
- Showed a strong less-is-more effect: training on ∼**3%** of AMASS can outperform the full corpus across trackers (e.g., Any2Track, TWIST2); validated cross-dataset transfer and real-world **Unitree G1** deployment without fine-tuning.

[4] **Humanoid-GPT: Scaling Data and Structure for Zero-Shot Motion Tracking**
Zekun Qi*, Xuchuan Chen*, Jilong Wang*, Chenghuai Lin*, Yunrui Lian, Zhikai Zhang, ***Dairu Liu***, Wenyao Zhang, Yu Guan, Xinqiang Yu, He Wang†, Li Yi†
CVPR 2026 [Project] [Code]

- Introduced **Humanoid-GPT**, a GPT style causal Transformer trained on a ∼**2B-frame** retargeted motion corpus, distilling hundreds of RL motion experts via DAgger into a single general motion tracker with strong zero-shot generalization on **Unitree G1**.

- Proposed **Harmonic Motion Embedding** (**HME**) for diversity-aware, distribution-balanced sampling; characterized scaling trends for data and model size and deployed with real-time inference (ONNX/TensorRT, $<$1.5 ms on RTX 4090).

[5] **Evaluating Time Awareness and Cross-modal Active Perception of Large Models via 4D Escape Room Task**
Yurui Dong*, Ziyue Wang*, Shuyun Lu*, ***Dairu Liu****, Xuechen Liu*, Fuwen Luo, Peng Li†, Yang Liu†
EMNLP 2026 Under Review [Paper] [Project]

- Developed **EscapeCraft-4D**, a customizable 4D escape-room benchmark (**66 scenes**, **1,032 objects**) incorporating trigger-based audio, misleading auditory distractors, and temporally transient clues for evaluating time-aware multimodal reasoning.
- Introduced 4D specific process level metrics beyond final success rate: **AMR** (Audio Misleading Rate, measuring how often agents are misled by auditory distractors) and **TCSS** (Transient Clue Seeing Speed, measuring how early agents discover irreversibly-disappearing clues), revealing significant modality bias and failures in time-conditioned decision making across Omni models.

RESEARCH EXPERIENCE

Research Intern, GALBOT | IIIS, Tsinghua University — Beijing, China
Advisor: **Prof. Li Yi** — *Dec. 2025 – Present*

- Built a large-scale humanoid motion tracking benchmark and proposed a preference-aligned reward model for whole-body control evaluation, addressing the misalignment between kinematic metrics and human perception. [1]

Research Intern, THUNLP | AIR, Tsinghua University — Beijing, China
Advisor: **Prof. Peng Li**, **Prof. Yang Liu** — *Apr. 2024 – Present*

- Proposed Visual Abstract Thinking, a novel multimodal reasoning paradigm that enhances MLLMs via visual abstraction rather than verbose verbal rationales. [2]
- Developed a 4D escape-room benchmark for evaluating time awareness and cross-modal active perception in Omni models. [5]
- Exploring self-evolving agents that generalize from weak to strong supervision.

HONOURS AND AWARDS

National Scholarship (Top 0.2%; Undergraduate, 10000CNY) — Sep. 2024
Second Prize, NOIP 2018 (National Olympiad in Informatics in Provinces) — Dec. 2018
Second Prize, CSP-S 2019 (CCF Certified Software Professional) — Dec. 2019

LEADERSHIP AND SERVICES

Vice President, Quantitative Finance & Investment Society, Nankai University — Sep. 2025 – Present
Member, University Ski Team, Nankai University — 2024 – 2025
Academic Department Vice-Head, Psychology Association, Nankai University — Sep. 2024 – Sep. 2025

SKILLS

Robot Platforms: Unitree G1
Programming: C++, Python, MATLAB, LaTeX
Languages: Mandarin Chinese (native); English (IELTS 7.0 – L 7.5, R 8.0, S 6.0, W 6.0; CET-6 608)
Interests: Skiing